Marking Rubric for Lab #3 Assessment (Week 12)

FIT3143 Parallel Computing © Monash University 2026

LAB3 Marking Rubric

		HD	D	C	P	N
Methodology (50%)						
Task 1 **EV Charging Nodes**	Correctness, Performance, and Clarity (15%)	The program is capable of simulating an EV charging node with the correct MPI virtual topology. The program is capable of simulating charging ports correctly with OpenMP/POSIX threads and a shared array, without any noticeable errors. The program is capable of performing correct communication and associated computations to handle EV charging congestion and base station reporting according to the specification, without any noticeable errors. Clear diagrams and codes (or pseudo-codes) are used to present the architecture and implementation. The report sent to the base station contains extensive valuable information and complies with the requirement in the specification. The program can shutdown gracefully according to the specification requirements	The program is capable of simulating an EV charging node with the correct MPI virtual topology. The program is capable of simulating charging ports correctly with OpenMP/POSIX threads and a shared array, without any noticeable errors. The program is capable of performing correct communication and associated computations to handle EV charging congestion and base station reporting according to the specification, without any noticeable errors. Clear diagrams and codes (or pseudo-codes) are used to present the architecture and implementation. The report sent to the base station contains extensive valuable information and complies with the requirement in the specification. The program can shutdown gracefully according to the specification requirements	The program is capable of simulating an EV charging node with the correct MPI virtual topology. The program is capable of simulating charging ports correctly with OpenMP/POSIX threads and a shared array, without any noticeable errors. The program is capable of performing correct communication and associated computations to handle EV charging congestion and base station reporting according to the specification, without any noticeable errors. Clear diagrams and codes (or pseudo-codes) are used to present the architecture and implementation. The report sent to the base station contains extensive valuable information and complies with the requirement in the specification. The program can shutdown according to the specification requirements,	The program is capable of simulating an EV charging node with the correct MPI virtual topology. The program is capable of simulating charging ports correctly with OpenMP/POSIX threads and a shared array, despite some minor errors. The program is capable of performing correct communication and associated computations to handle EV charging congestion and base station reporting according to the specification, despite some minor errors. Diagrams and codes (or pseudo-codes) are used to present the architecture and implementation.	Task 1 is missing in the presentation, or missing in the submitted presentation material (or) The code contains major flaws affecting the correctness of the program (or) The code completely fails to perform parallelism using Open MPI or did not use Open MPI (or) The presentation contains major logical flaws indicating major conceptual errors (or) The code does not implement and use MPI virtual topologies (or) The program fails to be properly executed due to major bugs/flaws (including deadlock and system crash)

		and without any errors. Note: External user interruption (e.g., CTRL + C, or sending a 'kill -9 <pid>' command by user) is not allowed. Extensive optimisation has been made to improve computational and communicational efficiency, with strong and convincing experimental results (in tables and graphs) justifying the improvements. Multiple versions (including both blocking and nonblocking MPI methods) have been implemented, explored, examined, and experimented to determine the best approach.	and without any errors. Note: External user interruption (e.g., CTRL + C, or sending a 'kill -9 <pid>' command by user) is not allowed. Significant optimisation has been made to improve computational and communicational efficiency, with experimental results (in tables and graphs) justifying the improvements.	despite system errors may occur. Note: External user interruption (e.g., CTRL + C, or sending a 'kill -9 <pid>' command by user) is not allowed.		
Task 2 **Base station**	Correctness, Performance, and Clarity (15%)	The program is capable of simulating the Base Station with a single MPI process. The program is capable of periodically listening and reading incoming reports from the EV charging nodes. The program is capable of reporting and communicating all nearest available neighbouring nodes correctly according to the specification, without any noticeable errors. The program is capable of logging key performance metrics to an output file according to the specification, without any noticeable errors and omissions.	The program is capable of simulating the Base Station with a single MPI process. The program is capable of periodically listening and reading incoming reports from the EV charging nodes. The program is capable of reporting and communicating all nearest available neighbouring nodes correctly according to the specification, without any noticeable errors. The program is capable of logging key performance metrics to an output file according to the specification, without any noticeable errors and omissions.	The program is capable of simulating the Base Station with a single MPI process. The program is capable of periodically listening and reading incoming reports from the EV charging nodes. The program is capable of reporting and communicating all nearest available neighbouring nodes correctly according to the specification, without any noticeable errors. The program is capable of logging key performance metrics to an output file according to the specification, without any noticeable errors and omissions.	The program is capable of simulating the Base Station with a single MPI process. The program is capable of periodically listening and reading incoming reports from the EV charging nodes. The program is capable of reporting and communicating (at least 1) nearest available neighbouring nodes correctly according to the specification, despite minor errors. The program is capable of logging key performance metrics to an output file according to the specification, despite minor errors/omissions.	Task 2 is missing in the presentation, or missing in the submitted presentation material (or) The code contains major flaws affecting the correctness of the program (or) The code completely fails to perform parallelism using Open MPI or did not use Open MPI (or) The presentation contains major logical flaws indicating major conceptual errors (or) More than 1 MPI process has been

		The program can send a termination message to all EV charging nodes to gracefully shutdown the program without any errors. Note: External user interruption (e.g., CTRL + C, or sending a 'kill -9 <pid>' command by user) is not allowed. Clear diagrams and codes (or pseudo-codes) are used to present the architecture and implementation. Extensive optimisation has been made to improve computational and communicational efficiency, with strong and convincing experimental results (in tables and graphs) justifying the improvements. At least three versions (including both blocking and nonblocking MPI methods) have been implemented, explored, examined, and experimented to determine the best approach.	The program can send a termination message to all EV charging nodes to gracefully shutdown the program without any errors. Note: External user interruption (e.g., CTRL + C, or sending a 'kill -9 <pid>' command by user) is not allowed. Clear diagrams and codes (or pseudo-codes) are used to present the architecture and implementation. Significant optimisation has been made to improve computational and communicational efficiency, with experimental results (in tables and graphs) justifying the improvements.	The program can send a termination message to all EV charging nodes to shutdown the program, despite system errors may occur. Note: External user interruption (e.g., CTRL + C, or sending a 'kill -9 <pid>' command by user) is not allowed. Clear diagrams and codes (or pseudo-codes) are used to present the architecture and implementation.	Diagrams and codes (or pseudo-codes) are used to present the architecture and implementation.	created/used by the Base Station (or) The program fails to be properly executed due to major bugs/flaws (including deadlock and system crash)
Task 3 Analysis	Correctness, Performance, and Clarity (20%)	Capable of correctly measuring communication time and compute communication delay for the implementation in Task 1 and Task 2. Communication delay on multiple (at least 30 different) EV grid sizes has been reported in table(s)/graph(s). Communication delay on multiple (at least 30	Capable of correctly measuring communication time and compute communication delay for the implementation in Task 1 and Task 2. Communication delay on multiple (at least 20 different) EV grid sizes has been reported in table(s)/graph(s). Communication delay on multiple (at least 20	Capable of correctly measuring communication time and compute communication delay for the implementation in Task 1 and Task 2. Communication delay on multiple (at least 5 different) EV grid sizes has been reported in graph(s). Communication delay on multiple (5 different) line topology instances has been	Capable of correctly measuring communication time and compute communication delay for the implementation in Task 1 and Task 2, despite some minor errors can be found. Communication delay is reported.	Task 3 is missing (or) Communication analysis is missing (or) Failed to measure communication time and compute communication delay

		different) line topology instances has been reported in table(s)/graph(s). Answers to the two questions in Task 3 are convincing and correct. Evidence is strong and there are no data omissions. All the required tables/graphs/diagrams are clear and readable, without any data omissions/errors. CAAS (or local cluster) has been used extensively in all the results to simulate physical environment and communication delay.	different) line topology instances has been reported in table(s)/graph(s). Answers to the two questions in Task 3 are convincing and correct. All the required tables/graphs/diagrams are clear and readable. CAAS (or local cluster) has been used extensively in all the results to simulate physical environment and communication delay.	reported in graph(s). Answers to the two questions in Task 3 are largely convincing and correct. All the required tables/graphs/diagrams are largely clear and readable.		
Presentation Quality and Q&A (50%)						
Presentation Quality (Clarity and Organisation) (10%)		The presentation is between 6 to 7 minutes. The presentation is <u>exceptionally</u> clear, well-organised, and easy to follow. The presentation discusses major issues, limitations, and future work <u>well</u>. The presentation is concise, precise, and accurate (i.e., materials have been digested, analysed, and chosen carefully without unnecessary information or details). Students are required to extensively use parallel	The presentation is between 6 to 7 minutes. The presentation is clear and organised with minor areas for improvement. The presentation discusses major issues, limitations, and future work.	The presentation is generally understandable but has major rooms for improvements.	The presentation is somewhat understandable but lacks either organisation or content flow.	The presentation is largely unclear, poorly organised, and hard to follow.

	computing terminologies (instead of general computing terms) at this level.				
Q&A (40%)	Capable of understanding and answering all the questions without external tools. All the answers are correct without any flaws. The answers are concise, precise, and accurate (i.e., fully answer the question without unnecessary information or details). Students are required to answer by using parallel computing terminologies (instead of general computing terms) at this level.	Capable of understanding all the questions and answering all the questions without external tools. The answers are all correct without any flaws.	Capable of understanding all the questions and answering all the questions without external tools. The answers are largely correct, despite some minor flaws.	Capable of understanding all the questions and answering all the questions without external tools. The answers are largely correct, despite some major flaws.	Fails to understand the questions (or) Relies on external tools (e.g., AI, software tools, or online tools) to understand the questions or answer the questions. (or) Answers are completely wrong with major flaws/issues.

Special Consideration & Extensions

- From the special consideration procedure, short extensions are <u>not available</u> for group assessments, in-class assessments (including in-class tests/quizzes and presentations), practical assessments (including laboratories), mid-semester tests, or scheduled final assessments. For these assessment types, students must instead apply for special consideration.
- Unless a special consideration extension is approved, the student group/team will generally receive a 0 mark if they fail to attend the in-class session, fail to demonstrate the work, or fail to submit the required files to Moodle.
- For students with special considerations approved by the special consideration team, an alternative form of assessment must be submitted no later than the newly approved deadline. Students are reminded to read and follow the special consideration and alternate assessment sections in the assessment specification.
- If the extended due date falls after Week 14 and you plan to submit on/after Week 14, students are advised to contact the Chief Examiner and/or the Unit Coordinator, as grades may need to be withheld to allow extra time for marking.
- Students who have received special considerations but failed to submit the required files to Moodle by the newly approved deadline for their alternative assessment will be treated as having made no submission (or as having submitted late, if the file eventually arrives after the newly approved due date). For alternative assessments, late penalties will still apply for late submissions.